LazerGamebox: outdoor game box, ESP32 + UHF radio + LoRa + GNSS

2-layer 70 x 100 mm, JLCPCB assembled. Hierarchical sheets below; nets cross sheets as global labels, rails as power symbols.

power

USB-C, TP4056 charger, DW01A/FS8205A protection,
load sharing, TPS62A02 3V3 buck, TPS61023 radio
boost, P-FET load switches, VBAT sense

File: power.kicad_sch

mcu

ESP32-WROOM-32E-N16, CH340C USB serial with
auto-reset, LIS3DHTR accelerometer,
TCA9555 I2C GPIO expander

File: mcu.kicad_sch

radio

SA818S-U 1 W UHF module, PTT driver,
PD divider, H/L jumper, pi pads, SMA

File: radio.kicad_sch

lora_gps

Ra-01SH LoRa 868 MHz, ATGM336H GNSS + patch,
SIM7080G cellular option (DNP) with SIM and
1.8 V level shifters

File: lora_gps.kicad_sch

audio

TS5A3159 source mux, NS4150B class-D + speaker,
DAC to radio mic divider, MEMS mic + LMV321,
radio AF or mic to ADC via mux

File: audio.kicad_sch

io

2.0in ST7789 display on 12p ZIF + backlight,
10k pot, 3 buttons, 3x WS2812B-2020,
IR receiver + high-power IR emitter

File: io.kicad_sch

MECHANICAL: 4x M2.5 mounting holes, 3 fiducials

H1 M2.5

H2 M2.5

H3 M2.5

H4 M2.5

FID1 Fiducial

FID2 Fiducial

FID3 Fiducial

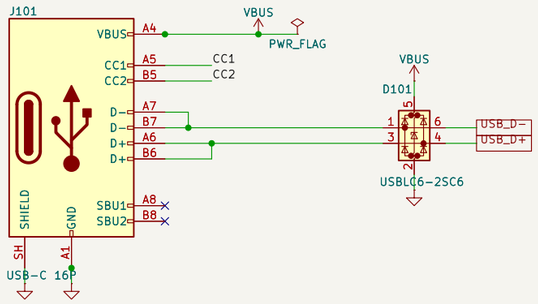
Cross-sheet signal list: see docs/pinmap.md. Power rails: VBUS, VBAT, VSYS, +3V3, RADIO_4V5, VBAT_SW, GPS_3V3, LCD_3V3, GND.

Sheet: /	
File: LazerGamebox.kicad_sch	
Title: LazerGamebox	
Size: A3	Date:
KiCad E.D.A. 10.0.3	Rev: A
	Id: 1/7

POWER: USB-C PD sink, BQ25601 charger with power path, 2x 18650, fuel gauge, 3V3 buck, radio boost, load switches

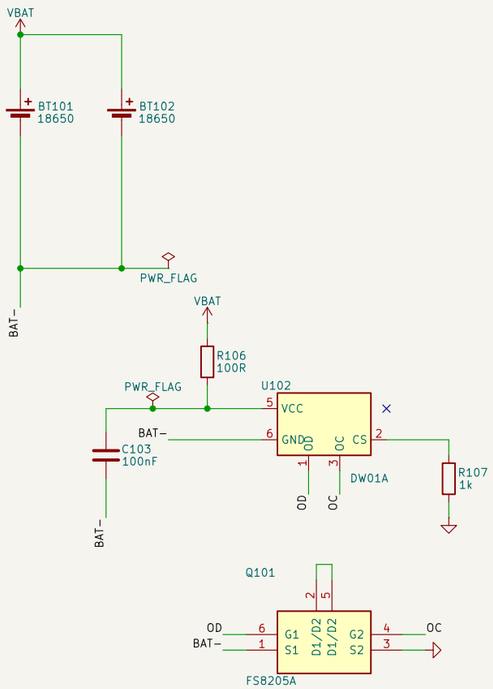
All switched rails default OFF at power-up (gate pull-downs); firmware enables them through the TCA9555 expander.

USB-C INPUT



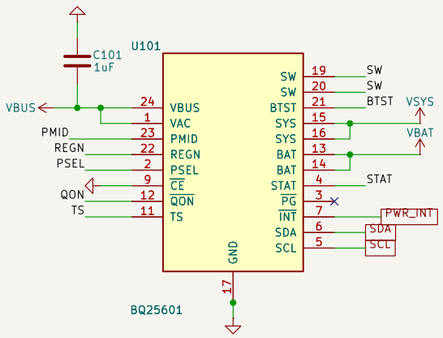
CC1/CC2 go to the HUSB238 PD sink (Rd inside it). D+/D- both rows joined (reversible plug).
VBUS: 5 V only, up to 3 A negotiated; USBLC6 clamps VBUS and the data lines.

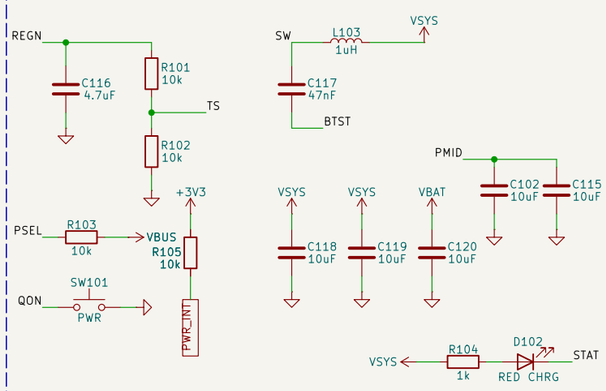
2x 18650 IN PARALLEL + PROTECTION (DW01A + FS8205A)



Two cells of the same type and age, fitted together (parallel pair, no balancing needed).
Cell negative reaches system GND through the dual MOSFET (BAT- net).
DW01A: 2.5 V under-voltage, 4.3 V over-voltage,
150 mV over-current sense (2 to 4 A with the FS8205A).

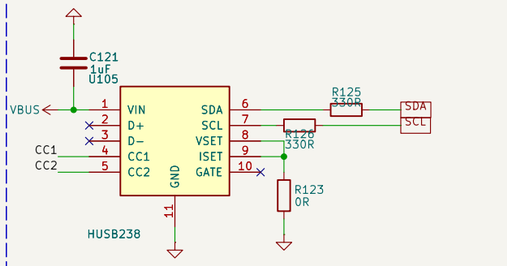
CHARGER + POWER PATH (BQ25601, 2 A, I2C 0x6B)

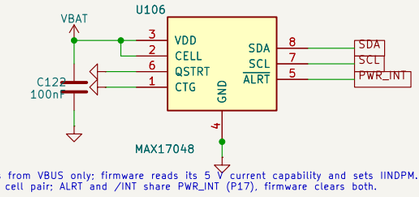




NVDC power path: VSYS runs from USB or the cells; BATFET ship mode = true off.
Plug-in defaults: 500 mA input (PSEL high), 2 A charge, 4.208 V. Firmware raises
IINDPM after reading the HUSB238, clears EN_HIZ after a QON wake, feeds or
disables the 40 s watchdog.

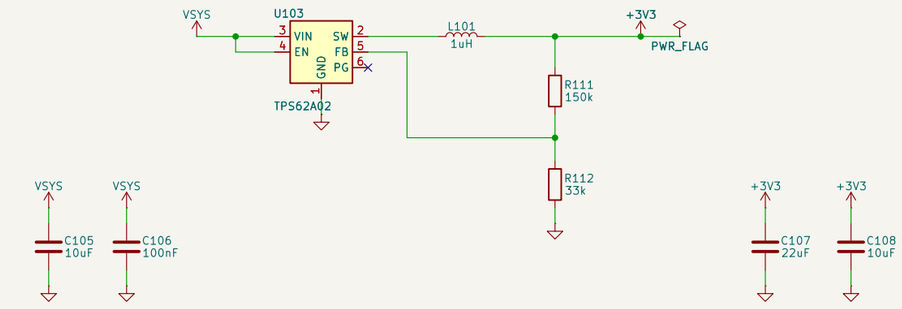
PD SINK (HUSB238 0x08) + GAUGE (MAX17048 0x36)





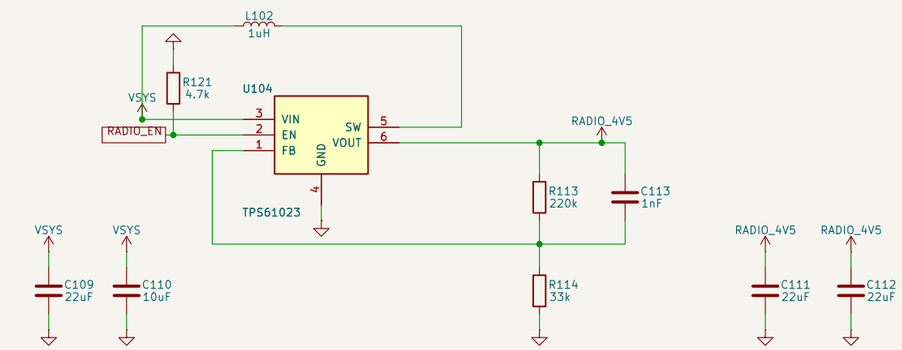
HUSB238 runs from VBUS only; firmware reads its 5 V current capability and sets IINDPM.
Gauge on the cell pair; ALRT and /INT share PWR_INT (P17), firmware clears both.

3V3 BUCK, ALWAYS ON (TPS62A02, 2 A, Iq 23 uA)



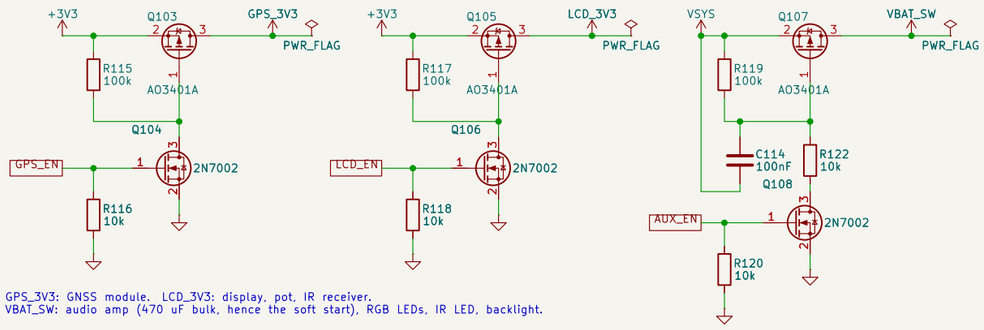
VOUT = 0.6 V x (1 + 100k/22k) = 3.33 V. L = 1 uH (SWPA4030S1R0, 5.7 A sat). fsw 2.4 MHz.
EN tied to VIN: rail is always on; deep-sleep current is set by the ESP32 and the buck Iq.

RADIO SUPPLY: 4.6 V BOOST WITH TRUE DISCONNECT (TPS61023)



VOUT = 0.6 V x (1 + 220k/33k) = 4.6 V. Pass-through when the cell is above 4.6 V.
EN low = output disconnected (< 1 uA): no separate load switch for the radio. 470 uF bulk sits at the module (radio sheet).

LOAD SWITCHES (P-FET high side, N-FET level shift, default OFF)



GPS_3V3: GNSS module. LCD_3V3: display, pot, IR receiver.
VBAT_SW: audio amp (470 uF bulk, hence the soft start), RGB LEDs, IR LED, backlight.

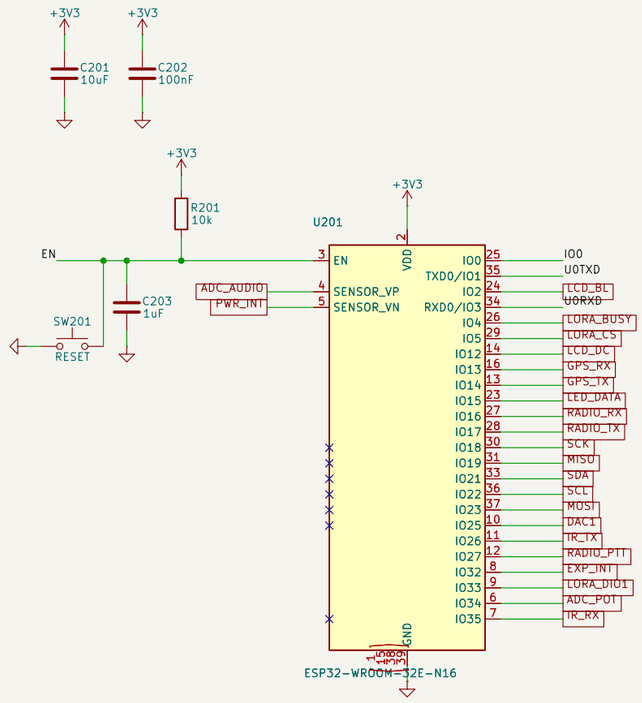
PWR_FLAG on GND (ERC only).

Sheet: /power/		
File: power.kicad_sch		
Title: power		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 2/7

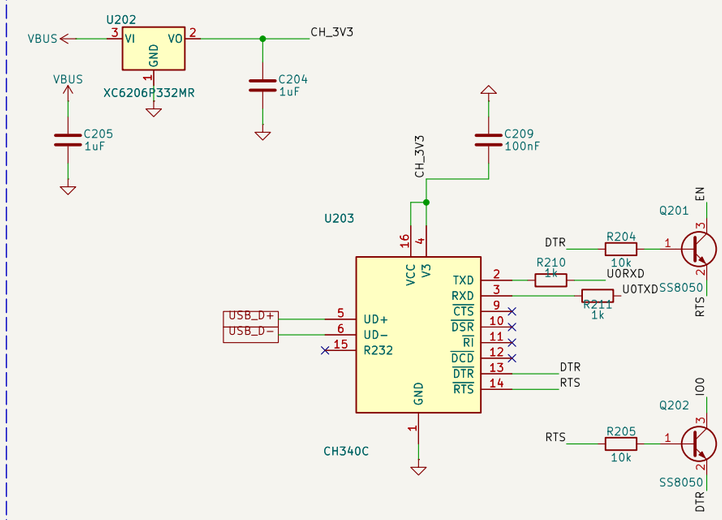
MCU: ESP32-WROOM-32E, USB serial (CH340C on its own 3.3 V LDO), accelerometer, I2C GPIO expander

All 26 usable GPIO are used; slow enables, resets and buttons live on the TCA9555 (I2C 0x20). See docs/pinmap.md.

ESP32-WROOM-32E-N16 (16 MB flash)

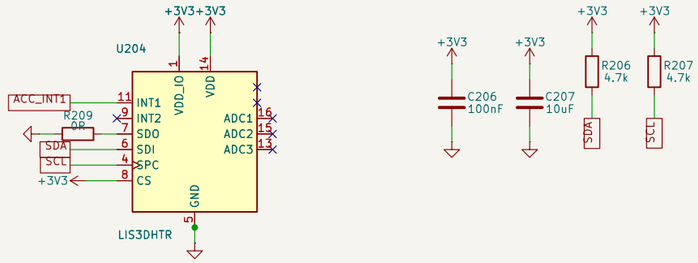


USB SERIAL: CH340C + auto-reset



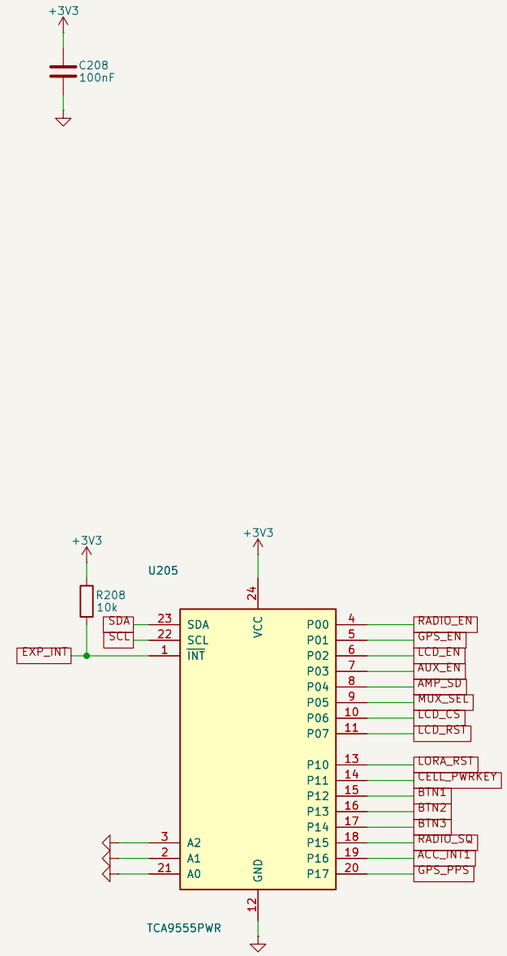
CH340C runs from its own 3.3 V LDO on VBUS (V3 tied to VCC): 3.3 V logic into the ESP32, unpowered on battery. DTR/RTS transistor pair = esptool auto-reset.

ACCELEROMETER (LIS3DHTR, I2C 0x18)



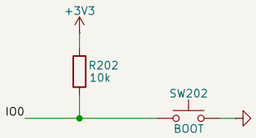
SA0 (SDO) pulled low by R209 = address 0x18, CS high = I2C mode. INT1 wakes the ESP32 via the expander INT.

I2C GPIO EXPANDER (TCA9555, 0x20)

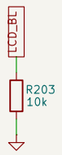


P00–P11 outputs, P12–P17 inputs (100k internal pull-ups). INT open drain -> GPIO32, ext0 wake source.

BOOT / RESET BUTTONS



IO0 low at reset = download mode. RESET pulls EN low.
Strapping: IO2 has a 10k pull-down (R203, net LCD_BL), IO12 has no pull-up (flash stays 3.3 V), IO5 and IO15 keep their internal pull-ups.

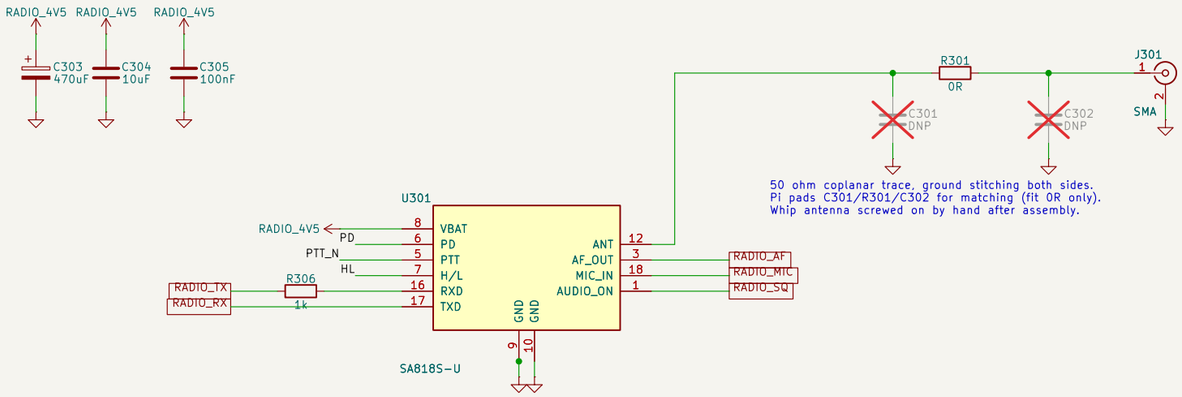


Sheet: /mcu/	
File: mcu.kicad_sch	
Title: mcu	
Size: A3	Date:
KiCad E.D.A. 10.0.3	**Rev: A**
	Id: 3/7

RADIO: SA818S-U 1 W UHF transceiver, 400-480 MHz

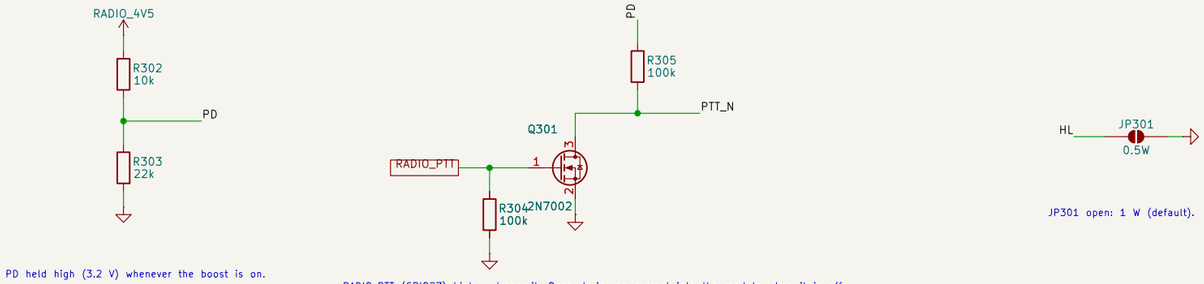
UART2 9600 8N1 AT commands. Supply 4.6 V from the boost, RADIO_EN low = module fully off.

SA818S-U MODULE



Module current: RX 60 mA, TX high 750 mA, TX low 450 mA, 1 uA in power-down. 470 uF right at the VBAT pad.
H/L open = 1 W; bridge JP301 for 0.5 W (the pin must never be driven high).

MODULE CONTROL



PD held high (3.2 V) whenever the boost is on.

RADIO_PTT (GPIO27) high = transmit. Open drain: no current into the module when it is off.
R305 pulls PTT to the PD node (3.2 V, follows the boost rail): RX by default, nothing back-drives a dead module.

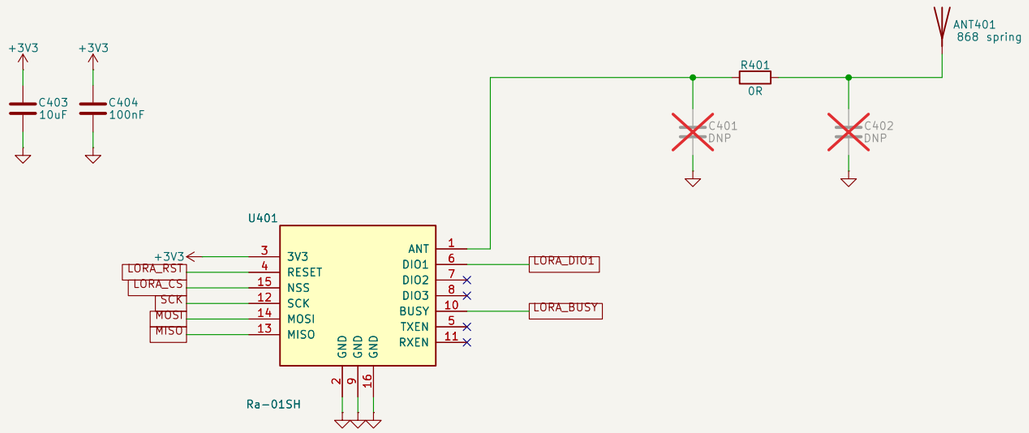
JP301 open: 1 W (default).

Sheet: /radio/		
File: radio.kicad_sch		
Title: radio		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 4/7

LoRa 868 MHz (Ra-01SH), GNSS (ATGM336H-5N31), optional cellular (SIM7080G, all DNP)

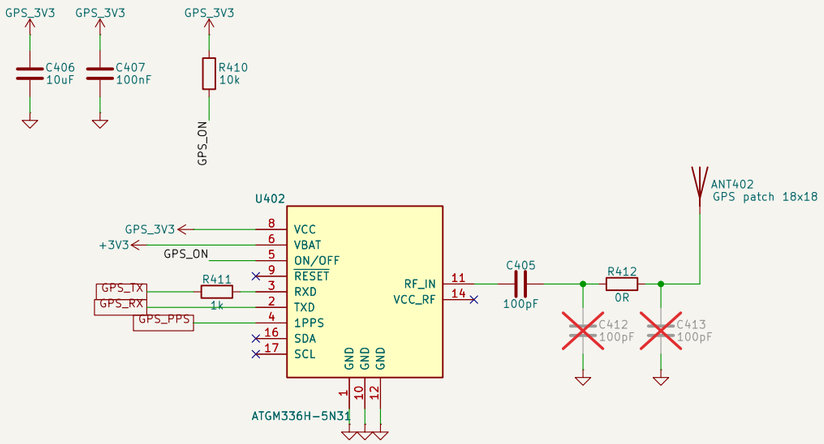
LoRa on VSPI with its own CS. GPS on UART1 behind GPS_3V3. Cellular shares UART1 through 0R links and needs 1.8 V level shifting.

LoRa Ra-01SH (SX1262)



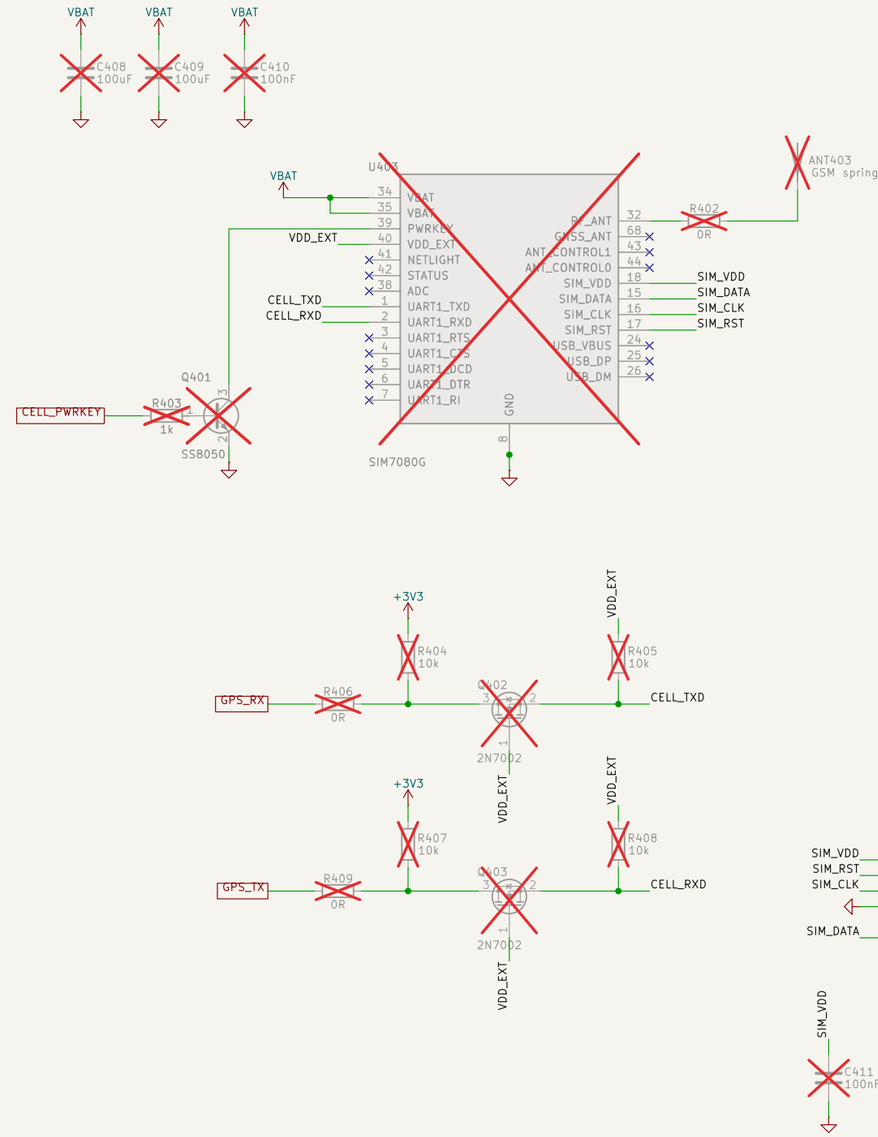
RF switch is driven by DIO2 inside the module: setDio2AsRfSwitch(true). TXEN/RXEN pads unused.
SX1262 sleep 1 uA, always powered from +3V3. 50 ohm trace to the spring antenna, keep 15 mm from the UHF PA.

GNSS ATGM336H-5N31 (UART1, 1PPS)



Passive patch antenna on a solid ground pour (40 x 40 mm), 15 mm or more from the UHF PA and the LoRa antenna.
VBAT backup pin on +3V3 (10 uA) keeps ephemeris for hot starts while GPS_3V3 is switched off. ON/OFF pulled up by R410.

CELLULAR OPTION: SIM7080G Cat-M / NB-IoT + nano-SIM + antenna (ALL DNP)



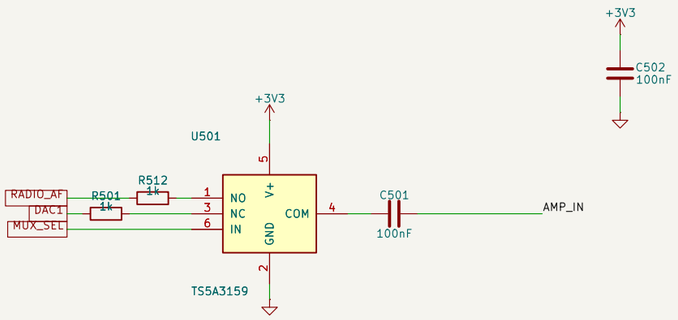
Fit EITHER the GNSS module OR the cellular module (SIM7080G has its own GNSS): both share UART1 via R406/R409.
SIM7080G IO is 1.8 V (VDD_EXT): Q402/Q403 are BSS138-style bidirectional level shifters.
PWRKEY pulsed low 1 s by CELL_PWRKEY through Q401. No ESD array on the SIM lines: socket sits inside the enclosure.

Sheet: /lora_gps/		
File: lora_gps.kicad_sch		
Title: lora_gps		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 5/7

AUDIO: DAC or radio audio -> class-D amp -> speaker; DAC and mic -> radio mic; radio audio or mic -> ADC

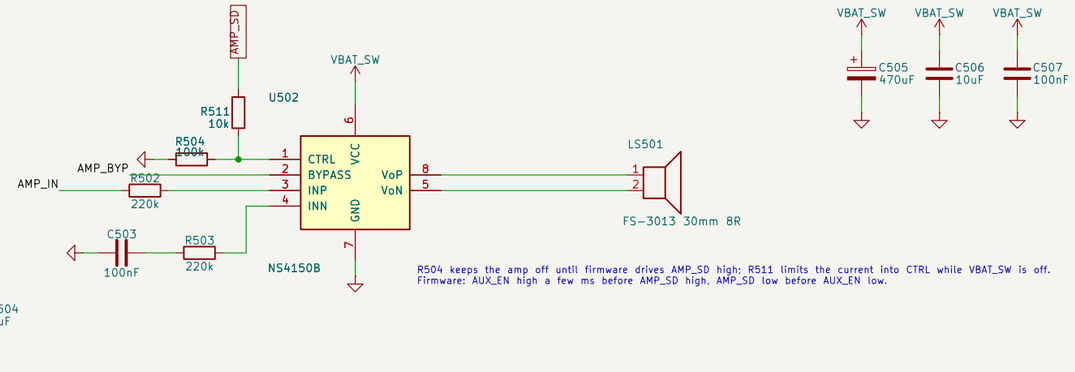
NS4150B gain = 2 x 240k / 220k = 2.2. MUX_SEL low = ESP32 DAC to the speaker, high = radio receive audio to the speaker.

SOURCE SELECT (TS5A3159)



DAC1 is 0–3.3 V unipolar; R501 limits current into the switch.
AC coupling after the switch (C501).

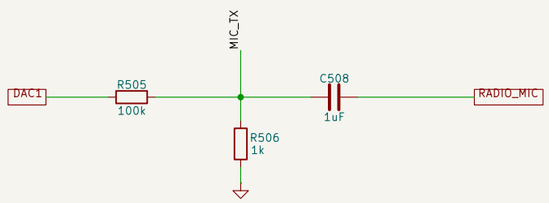
CLASS-D AMPLIFIER (NS4150B) + 30 mm SPEAKER



R504 keeps the amp off until firmware drives AMP_SD high; R511 limits the current into CTRL while VBAT_SW is off.
Firmware: AUX_EN high a few ms before AMP_SD high, AMP_SD low before AUX_EN low.

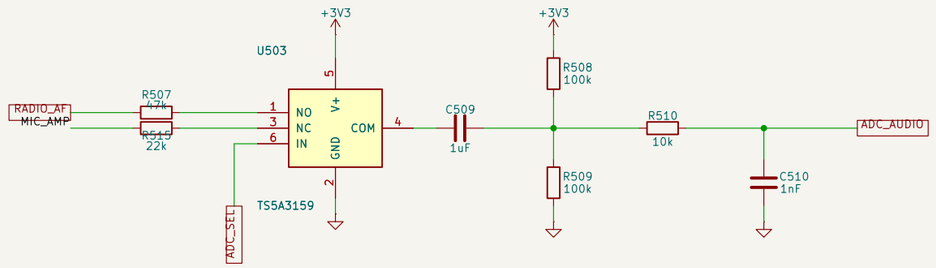
Filter-free BTL output: keep speaker traces short and away from the radio. VCC on VBAT_SW with 470 uF bulk.
Max output about 1 W into 8 ohm at 3.7 V.

DAC -> RADIO MIC_IN (100:1)



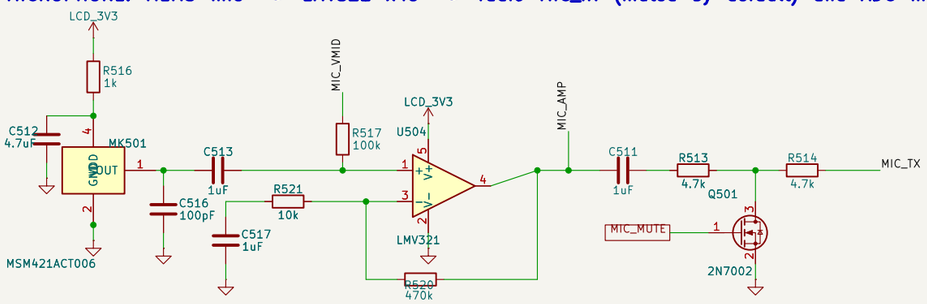
3.3 Vpp DAC -> 33 mVpp at MIC_IN, AC coupled; MIC_TX adds the
microphone (about 9 mVrms speech). DTMF, tones and voice over the radio.

ADC INPUT MUX (TS5A3159): RADIO AF_OUT or MIC -> ADC1 (software Goertzel DTMF decoder)

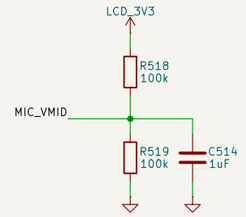


AF_OUT is 700 mV / 200 ohm: R507 with the 50k bias gives about 0.35 Vrms swing around 1.65 V at GPIO36; R515 passes 0.7 of MIC_AMP.
ADC_SEL high (expander default) = radio, low = microphone. R510/C510 = 16 kHz anti-alias for a 40 kHz ADC sample rate.

MICROPHONE: MEMS mic -> LMV321 x48 -> radio MIC_IN (muted by default) and ADC mux



MSM421ACT006: -42 dBV/Pa, 170 ohm output, 1.5 to 3.6 V, top port (hole in the enclosure above it). Runs on LCD_3V3 through R516/C512.
U504 gain 1 + R520/R521 = 48 (34 dB), bandwidth about 20 kHz from the 1 MHz GBW, output biased at LCD_3V3 / 2 (MIC_VMID).
Speech at 20 cm gives about 100 mVrms at MIC_AMP: R513/R514 into the 1k DAC node give about 9 mVrms at MIC_IN (rated deviation).
Q501 shorts the radio path unless MIC_MUTE is low (expander default high = muted). ADC path: R515 into U503 with ADC_SEL low.
Off in sleep with LCD_3V3 (about 0.3 mA when on). Firmware mutes the speaker while the mic is live or the radio transmits.

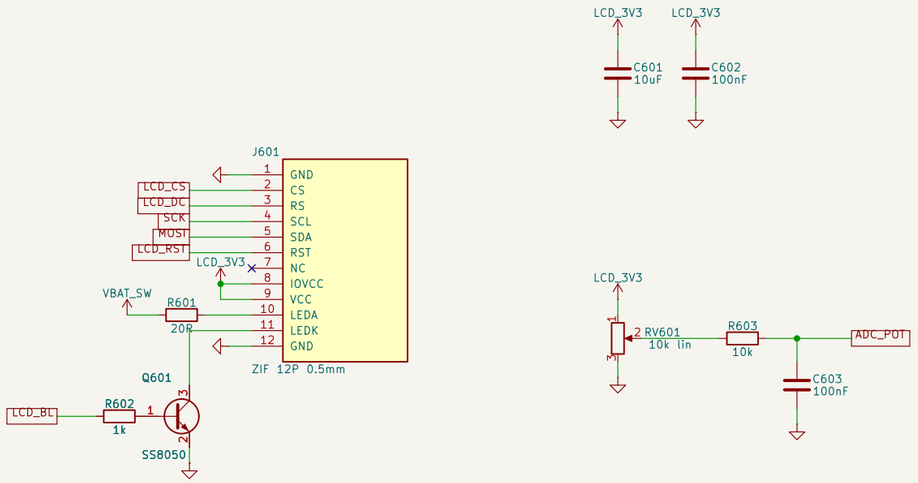


Sheet: /audio/	
File: audio.kicad_sch	
Title: audio	
Size: A3	Date:
KiCad E.D.A. 10.0.3	Id: 6/7
	Rev: A

IO: 2.0in display on ZIF, pot, buttons, RGB LEDs, IR transmit / receive

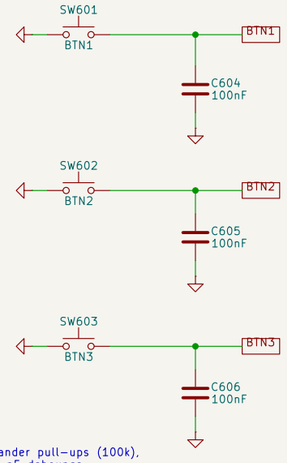
Front side of the board: display, buttons, pot, LEDs, IR. Display, pot and IR receiver ride on LCD_3V3.

DISPLAY: HS20HS072RX 2.0in IPS 320x240, ST7789V2, 12-pin 0.5 mm ZIF



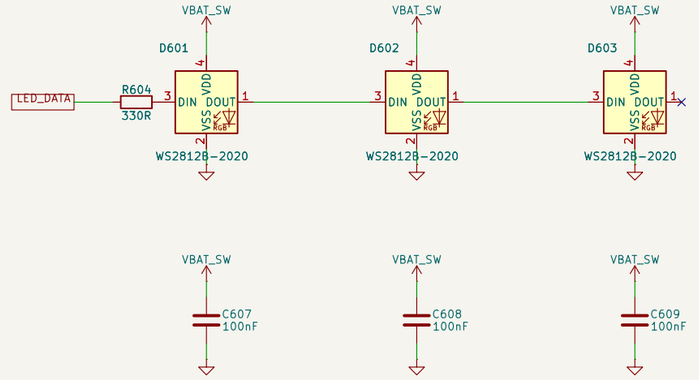
Backlight: 3 white LEDs in parallel from VBAT_SW via 20 R (25–60 mA over the cell range), PWM on LCD_BL.
Panel logic 3.3 V. Pot wiper -> GPIO34 (ADC1_CH6).

USER BUTTONS (to expander)



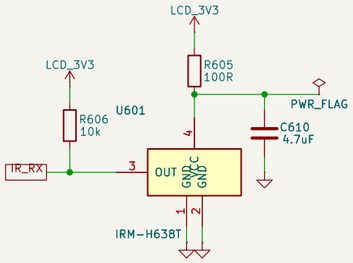
Expander pull-ups (100k),
100 nF debounce.

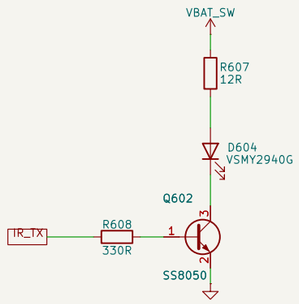
RGB LEDs (WS2812B-2020 x3)



LEDs run from VBAT_SW (3.0–4.2 V): data is 3.3 V, fine for WS2812B-V6.

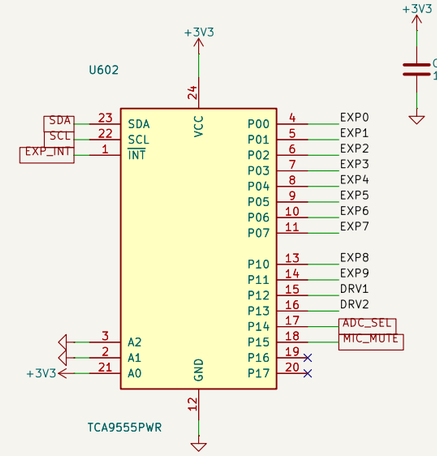
IR: 38 kHz RECEIVER and 940 nm HIGH-POWER EMITTER (laser-tag style)

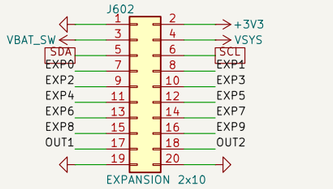


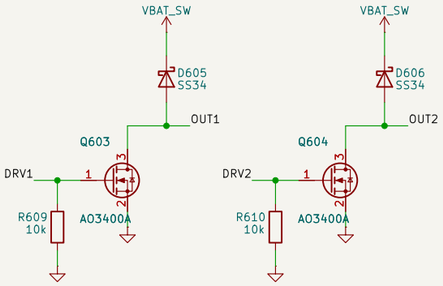


Receiver on LCD_3V3 (1.2 mA), output pulled up to 3.3 V -> GPIO35 (RMT). Never power it from VBAT_SW: its output would exceed 3.3 V.
Emitter: about 150–230 mA peak at 38 kHz / 50 % duty from VBAT_SW through 12 R (1206). Keep bursts under 100 ms.

EXPANSION (TCA9555 #2 at 0x21, 2x10 header, 2 drivers)







EXP0–9: 3.3 V GPIO, 10 mA sink, 25 mA source,
no PWM, about 1 ms latency.
OUT1/OUT2: open drain to GND, 2 A each; loads
return to VBAT_SW or +3V3. Gates pulled down,
off until firmware enables them.
I2C on the header for more gadgets.

Sheet: /io/	
File: io.kicad_sch	
Title: io	
Size: A3	Date:
KiCad E.D.A. 10.0.3	Id: 7/7